

INFINITE SIMPLE KAZHDAN GROUPS THAT ARE LIMITS OF FINITE SIMPLE GROUPS

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ABSTRACT. Infinite simple groups with property (T) can be limits of finite simple groups. For every infinite minimal subshift X , the group $G_X = \text{EL}_3(\text{LC}(X, \mathbb{F}_2) \rtimes \mathbb{Z})$ is an infinite, finitely generated, simple group with property (T) and a limit of finite simple groups $\text{SL}_{3N}(\mathbb{F}_2)$ whose Cayley graphs form expanders, so it is locally embeddable into finite groups (LEF), sofic and hyperlinear. This answers the question of Brown and Ozawa whether an infinite simple Kazhdan group can be hyperlinear, and Pestov's sofic version of it. A finitely generated group is LEF if and only if it is a subgroup of an infinite, finitely generated, simple Kazhdan group that is a limit of finite simple groups whose Cayley graphs form expanders, and the host can be chosen with a word problem of the same Turing degree as that of the group. One such host contains every recursively presented finitely generated LEF group.

Can an infinite simple group with property (T) be hyperlinear? Brown asked this in 2001 as a question about embeddings into unitary groups of McDuff factors that embed in an ultrapower $\mathcal{R}^\omega$ of the hyperfinite II_1 factor [5, §11, Question 7]. Ozawa stated the hyperlinear form in 2003, noting that the full C^* -algebra of such a group cannot have the local lifting property [31, §7], and Pestov asked the sofic form as Open question 9.1 [32]. Thom constructed a finitely generated Kazhdan group that is locally embeddable into finite groups but not residually finite, and so lacks the factorization property [38], but his example is not simple. Pestov and Kwiatkowska wrote that a hope of constructing groups that are not hyperlinear from property (T) remained, since it was unknown whether finitely generated simple Kazhdan groups could be hyperlinear [33]. Arzhantseva listed the sofic form among open questions, and gave an infinite simple Kazhdan group that is asymptotically residually finite, a weaker property in which the generating set may change with the radius [2, §3]. The groups below answer all three forms of the question positively.

Theorem 1. *Let $X \subseteq A^{\mathbb{Z}}$ be an infinite minimal subshift over a finite alphabet, with shift T , and let $\text{LC}(X, \mathbb{F}_2)$ be the ring of locally constant functions $X \rightarrow \mathbb{F}_2$. Then*

$$G_X = \text{EL}_3(\text{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z})$$

is an infinite, finitely generated, simple group with Kazhdan's property (T), and an expander limit, in the sense defined below, of finite simple groups $\text{SL}_{3N}(\mathbb{F}_2)$. So G_X is locally embeddable into finite groups (LEF), sofic and hyperlinear.

In Brown's form, G_X lies in the unitary group of the McDuff factor $L(G_X) \bar{\otimes} \mathcal{R}$, which embeds in $\mathcal{R}^\omega$ (Section 1).

Here X is a closed shift-invariant set in which every orbit is dense. A *marked group* is a group with a finite generating tuple. Marked groups $(Q_k, \bar{q}_k)$ converge to a marked group $(G, \bar{g})$ in the space of marked groups [14, 7] if the tuples have the same length m and, for each word

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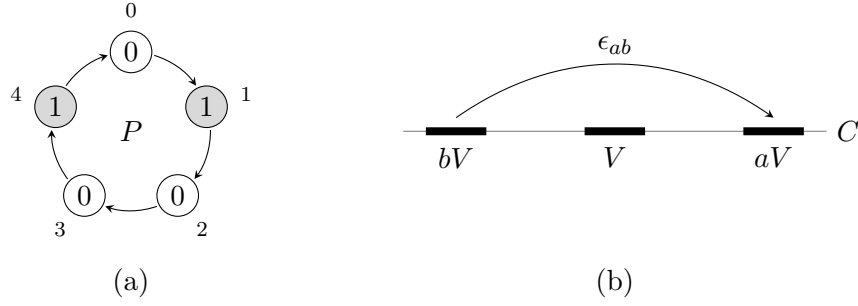


FIGURE 1. (a) The 5-periodic sequence 01001, which has the words of length 3 of the Fibonacci subshift, on a 5-cycle: u acts as the rotation P , and e_1 as the diagonal matrix with 1 at the shaded positions. (b) A small clopen set V and translates aV and bV ; the matrix units $\epsilon_{ab} = e_{aV}u_{ab^{-1}}$ move bV to aV .

w in m letters and their inverses and all large k depending on w , we have $w(\bar{q}_k) = 1$ if and only if $w(\bar{g}) = 1$. We call $(G, \bar{g})$ an *expander limit* of the $(Q_k, \bar{q}_k)$ if moreover the Q_k are finite, $|Q_k| \rightarrow \infty$, and their Cayley graphs with respect to the $\bar{q}_k$ form a family of expanders: there is $\varepsilon > 0$ such that for every k , every set of at most $|Q_k|/2$ elements of Q_k has at least ε times as many neighbors outside it as it has elements, the neighbors of an element being its products with the entries of $\bar{q}_k$ and their inverses. A group is an expander limit of groups from a class if some marking of it is an expander limit of marked groups from that class.

A group is LEF [40] if each finite subset Φ maps injectively into a finite group by a map that preserves every product gh with $g, h, gh \in \Phi$. Limits of finite groups in the space of marked groups are LEF. Sofic and hyperlinear groups are defined by approximately multiplicative maps into finite symmetric groups with the normalized Hamming distance, and into unitary groups with the normalized Hilbert–Schmidt distance, that keep nontrivial elements uniformly far from 1 [32]. Hyperlinear groups were named by Rădulescu [34], and sofic groups were introduced by Gromov [17] and named by Weiss [41].

The proof has three ingredients.

- *Property (T)*. The ring $\text{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z}$ consists of the finite sums $\sum_j f_j u^j$ with $f_j \in \text{LC}(X, \mathbb{F}_2)$, multiplied using $ufu^{-1} = f \circ T^{-1}$. It is finitely generated, so G_X has Kazhdan’s property (T) [21] by the theorem of Ershov and Jaikin-Zapirain [12].
- *Simplicity*. A nontrivial normal subgroup contains a nontrivial commutator lying in a copy of the finite simple group $\text{GL}_d(\mathbb{F}_2)$ over a clopen tower, a small clopen set with its disjoint translates (Figure 1(b)). So it contains this group and with it an elementary matrix, from which minimality gives every elementary matrix.
- *Finite models*. A periodic sequence of period N approximating X turns the shift into a cyclic permutation matrix and each locally constant function into a diagonal matrix (Figure 1(a)), as in the proof by Grigorchuk and Medynets that topological full groups of minimal Cantor systems are LEF [15, Theorem 2.6]. These models make G_X a limit of the groups $\text{SL}_{3N}(\mathbb{F}_2)$ in the space of marked groups. The $\text{SL}_{3N}(\mathbb{F}_2)$ are quotients of one Kazhdan group, EL_3 of a finitely generated free algebra over $\mathbb{F}_2$, so their Cayley graphs form expanders as in Kassabov’s construction [20], and G_X is an expander limit of them.

The same argument applies to other actions. An action on a Cantor set is topologically free if no nontrivial element fixes a nonempty open set. We write $e_{ij}(r) = I_n + rE_{ij}$, with E_{ij} the matrix unit, for the elementary matrices.

TABLE 1. Properties of G_X

Property	Reason	Where
finitely generated	by the $e_{ij}(s)$ with $s \in S$	§1, §2
infinite	$e_{12}(\text{LC}(X, \mathbb{F}_2))$ is infinite	§1
simple	small clopen sets carry $\text{GL}_d(\mathbb{F}_2)$; minimality	§1
property (T)	Ershov–Jaikin-Zapirain [12]	§1
expander limit of $\text{SL}_{3N}(\mathbb{F}_2)$	periodic approximations; [12, 25, 20]	§1, §2
LEF, sofic, hyperlinear	limit of finite groups; [32, 11]	§1
not residually finite	infinite and simple	§5
not finitely presented	finitely presented LEF groups are residually finite [37, 40]	§5
$L(G_X)$ is a II_1 factor in $\mathcal{R}^\omega$	infinite conjugacy classes; hyperlinear	§1
word problem	has the Turing degree of $L(X)$	Cor. 5

Theorem 2. *Let a finitely generated group Λ act minimally and topologically freely on a Cantor set C , and let the ring $R = \text{LC}(C, \mathbb{F}_2) \rtimes \Lambda$ be generated by a finite set $S \ni 1$.*

- (a) *For every $n \geq 3$, the group $\text{EL}_n(R)$ is an infinite, finitely generated, simple Kazhdan group.*
- (b) *Suppose that there are maps $\varphi_k: R \rightarrow M_{N_k}(\mathbb{F}_2)$ with $\varphi_k(1) = I$ such that $\varphi_k(S)$ generates the ring $M_{N_k}(\mathbb{F}_2)$, and for all $r, r' \in R$ and all large k , depending on r and r' , we have $\varphi_k(r + r') = \varphi_k(r) + \varphi_k(r')$ and $\varphi_k(rr') = \varphi_k(r)\varphi_k(r')$. Then for every $n \geq 3$, $\text{EL}_n(R)$, marked by the $e_{ij}(s)$ with $s \in S$, is an expander limit of the finite simple groups $\text{SL}_{nN_k}(\mathbb{F}_2)$ marked by the $e_{ij}(\varphi_k(s))$.*

Periodic approximations of X give such maps, and Theorem 1 follows. Lamplighter actions show that the countable subgroups of the groups in Theorem 2(b) are exactly the countable LEF groups (Corollary 4). Over $\mathbb{Z}$ the subshift hypothesis cannot be dropped (Section 2).

1. PROOF OF THEOREM 2

We write the proof for $n = 3$; for $n \geq 3$ replace 3 by n throughout.

The ring and property (T). Write e_U for the indicator of a clopen set $U \subseteq C$. The ring $R = \bigoplus_{\xi \in \Lambda} \text{LC}(C, \mathbb{F}_2) u_\xi$ has the multiplication $(f u_\xi)(f' u_\eta) = f(f' \circ \xi^{-1}) u_{\xi\eta}$, so $u_\xi e_U u_\xi^{-1} = e_{\xi U}$. Fix a finite symmetric generating set of Λ , let $|\xi|$ be the word length, and put $B_m = \{\xi \in \Lambda : |\xi| \leq m\}$. Let $G = \text{EL}_3(R)$ be the subgroup of $\text{GL}_3(R)$ generated by the $e_{ij}(r)$ with $i \neq j$ and $r \in R$. With $[g, h] = ghg^{-1}h^{-1}$ and i, j, l distinct,

$$(1) \quad e_{ij}(r + r') = e_{ij}(r)e_{ij}(r'), \quad [e_{il}(r), e_{lj}(r')] = e_{ij}(rr').$$

So the $e_{ij}(s)$ with $s \in S$ generate G . By Ershov and Jaikin-Zapirain [12, Theorem 1.1], EL_n of every finitely generated associative unital ring has property (T) for $n \geq 3$, so G has property (T). It is infinite because $e_{12}(\text{LC}(C, \mathbb{F}_2))$ is infinite.

Simplicity. Let $1 \neq K \trianglelefteq G$ and $1 \neq g \in K$. We show in three steps that $K = G$.

Step 1: a nontrivial commutator with a small elementary matrix. Let $w \geq 0$ bound the word lengths of the ξ with u_ξ occurring in the entries of g and g^{-1} . Call a clopen set V small if (i) $V \cap \xi V = \emptyset$ for every $\xi \in B_{2w} \setminus \{e\}$, and (ii) $f \circ \xi$ is constant on V for every $\xi \in B_w$ and every coefficient $f \in \text{LC}(C, \mathbb{F}_2)$ of an entry of g or g^{-1} . Each $\xi \neq e$ fixes a closed set with empty interior, so the points moved by every $\xi \in B_{2w} \setminus \{e\}$ form a dense open set. Small

enough clopen neighborhoods of these points are small, so every nonempty clopen set contains a nonempty small one.

Some $h = e_{ij}(e_V)$ with V small does not commute with g . Indeed, g commutes with $e_{ij}(r)$ exactly when $g_{pi}r = 0$ for $p \neq i$, $rg_{jq} = 0$ for $q \neq j$, and $g_{ii}r = rg_{jj}$. For $c = \sum_{\xi} c_{\xi}u_{\xi}$ and small V we have $ce_V = \sum_{\xi} c_{\xi}e_{\xi V}u_{\xi}$ and $e_Vc = \sum_{\xi} e_Vc_{\xi}u_{\xi}$. If $c_{\xi} \neq 0$, it equals 1 on a nonempty clopen set U , and a small set $V \subseteq \xi^{-1}U$ gives $ce_V \neq 0$, while a small set $V \subseteq U$ gives $e_Vc \neq 0$. So if g commutes with every such h , then g is diagonal. In $g_{ii}e_V = e_Vg_{jj}$ the coefficients at u_{ξ} with $\xi \neq e$ are supported in the disjoint sets ξV and V , so they vanish, and the same argument shows that the corresponding coefficients of g_{ii} and g_{jj} vanish. The coefficients at u_e agree on every small V , so they agree everywhere. Therefore $g = cI_3$ with $c \in \text{LC}(C, \mathbb{F}_2)$, and comparing coefficients at u_e in $cc^{-1} = 1$ gives $c = 1$ and $g = 1$. Fix such h and V . Then $x = [g, h]$ lies in $K \setminus \{1\}$.

Step 2: the commutator lies in a finite simple group over a tower. Put $d = 3|B_w|$. For $a, b \in B_w$ put $\epsilon_{ab} = e_{aV}u_{ab^{-1}}$ (Figure 1(b)). Since $\epsilon_{ab}\epsilon_{a'b'} = e_{a(V \cap b^{-1}a'V)}u_{ab^{-1}a'b'^{-1}}$ and $b^{-1}a' \in B_{2w}$, we get $\epsilon_{ab}\epsilon_{a'b'} = \delta_{ba'}\epsilon_{ab'}$. As $h \neq I_3$, V is nonempty, so the ϵ_{ab} span a ring $A_V \cong M_{|B_w|}(\mathbb{F}_2)$ with unit $1_V = \sum_a \epsilon_{aa}$, and $y \mapsto (1 - 1_V)I_3 + y$ embeds $\text{GL}_3(A_V) \cong \text{GL}_d(\mathbb{F}_2)$ in $\text{GL}_3(R)$. Its image H_V lies in G , since the transvections $e_{pq}(\epsilon_{ab})$ with $p \neq q$ generate it: the other transvections of $\text{GL}_d(\mathbb{F}_2)$ are commutators of these.

If $a, b \in B_w$ and f, f' are coefficients as above, then

$$fu_a e_V f'u_b = f e_{aV} (f' \circ a^{-1}) u_{ab} \in \{0, \epsilon_{a,b^{-1}}\},$$

since $f \circ a$ and f' are constant on V . The entries of $ghg^{-1} - I_3 = g e_V E_{ij} g^{-1}$ are sums of such products, and $h^{-1} = h = I_3 + \epsilon_{ee}E_{ij}$, so $x - I_3 = (ghg^{-1} - h)h$ and $x^{-1} - I_3 = h(ghg^{-1} - h)$ lie in $M_3(A_V)$. Then $y = 1_V I_3 + (x - I_3)$ and $y' = 1_V I_3 + (x^{-1} - I_3)$ satisfy $yy' = y'y = 1_V I_3$, so $x = (1 - 1_V)I_3 + y \in H_V$. As $K \cap H_V$ is a nontrivial normal subgroup of $H_V \cong \text{GL}_d(\mathbb{F}_2) = \text{PSL}_d(\mathbb{F}_2)$, which is simple as $d \geq 3$, we get $H_V \subseteq K$, and $e_{pq}(e_V) \in K$ for all $p \neq q$.

Step 3: K contains every elementary matrix. The set $J = \{r \in R : e_{pq}(r) \in K \text{ for all } p \neq q\}$ is a two-sided ideal by (1), since $e_{pq}(r'r) = [e_{pl}(r'), e_{lq}(r)]$ and $e_{pq}(rr') = [e_{pl}(r), e_{lq}(r')]$ for $l \notin \{p, q\}$. It contains e_V , so it contains every $e_{\xi V} = u_{\xi} e_V u_{\xi}^{-1}$. By minimality and compactness finitely many of these sets cover C , so $1 = 1 - \prod_i (1 - e_{\xi_i V}) \in J$. Then $J = R$ and $K = G$. This proves (a).

For derived topological full groups, Matui showed by a similar argument that a nontrivial normal subgroup meets a simple union of products of alternating groups on towers [26, Lemma 3.4 and Theorem 4.9]. For the ring $\text{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z}$ of Theorem 1, simplicity also follows from Stepanov's theorem on the normal structure of GL_n [36, Theorem 4.4]: this ring is simple [8, Corollary 4.6], and for any p, q in it there are r, r' , not both 0, with $pr + qr' = 0$, by a dimension count on a tower over one small cylinder.

Finite simple models. Let $\mathcal{A}$ be the free algebra over $\mathbb{F}_2$ on letters τ_s , $s \in S$, and let $\pi: \mathcal{A} \rightarrow R$ and $\rho_k: \mathcal{A} \rightarrow M_{N_k}(\mathbb{F}_2)$ be the ring homomorphisms with $\pi(\tau_s) = s$ and $\rho_k(\tau_s) = \varphi_k(s)$. Both are onto, so G and $\text{EL}_3(M_{N_k}(\mathbb{F}_2))$ are quotients of $\text{EL}_3(\mathcal{A})$. Here $\text{EL}_3(M_{N_k}(\mathbb{F}_2)) = \text{SL}_{3N_k}(\mathbb{F}_2)$, as transvections generate $\text{SL}_{3N_k}(\mathbb{F}_2)$, and each transvection inside one block is a commutator of transvections between blocks. The group $\text{EL}_3(\mathcal{A})$ is generated by the $e_{ij}(1)$ and $e_{ij}(\tau_s)$, and in both quotients $e_{ij}(1)$ and $e_{ij}(\tau_s)$ with $s = 1$ have the same image. A word of length λ or less in the $e_{ij}(\tau_s)$ has entries of degree at most λ in $\mathcal{A}$, and $\varphi_k \circ \pi = \rho_k$ on the finitely many such entries for large k , where $\varphi_k(0) = 0$ by additivity. So a word that is trivial in G is trivial in $\text{SL}_{3N_k}(\mathbb{F}_2)$ for large k . Conversely, let w be a word that is nontrivial in G . As G is simple,

$e_{12}(1)$ is a product of conjugates of w and w^{-1} in G . This relation holds in $\mathrm{SL}_{3N_k}(\mathbb{F}_2)$ for large k , where $e_{12}(1)$ maps to the transvection $e_{12}(I) \neq I$, so w is nontrivial in $\mathrm{SL}_{3N_k}(\mathbb{F}_2)$ for large k . Therefore these marked finite simple groups converge to G , so G is LEF, and $N_k \rightarrow \infty$ because G is infinite. Since $\mathrm{EL}_3(\mathcal{A})$ has property (T) [12, Theorem 1.1], the Cayley graphs of its finite quotients with respect to the images of its generators form a family of expanders [25, 20], so G is an expander limit of the $\mathrm{SL}_{3N_k}(\mathbb{F}_2)$. This proves (b). $\square$

Brown's formulation. Let G be a group as in Theorem 2(b). It is LEF, so it is sofic [32, Example 4.5] and hyperlinear [11, Theorem 2], and $L(G)$ embeds in $\mathcal{R}^\omega$ [23, 34], [31, Proposition 7.1]. As G is infinite and simple, its nontrivial conjugacy classes are infinite, so $L(G)$ is a II_1 factor. Then G lies in the unitary group of the McDuff factor $L(G) \bar{\otimes} \mathcal{R}$ [27], which embeds in $\mathcal{R}^\omega \bar{\otimes} \mathcal{R} \subseteq (\mathcal{R} \bar{\otimes} \mathcal{R})^\omega \cong \mathcal{R}^\omega$, as in Brown's question.

2. PROOF OF THEOREM 1

Let $\mathbb{Z}$ act on X by powers of T , where $(Tx)_t = x_{t+1}$, and write $u = u_1$, so $ufu^{-1} = f \circ T^{-1}$. Minimality and infiniteness imply that X has no isolated points and T has no periodic points, so X is a Cantor set and the action is topologically free. The ring $R = \mathrm{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z}$ is generated by $S = \{1, u, u^{-1}\} \cup \{e_a : a \in A\}$, where $e_a = e_{\{x: x_0=a\}}$, since products of the translates $u^j e_a u^{-j}$ are the indicators of the cylinders, the sets of points with a given word at given coordinates, which span $\mathrm{LC}(X, \mathbb{F}_2)$.

Fix $x \in X$ and $\ell \geq 0$. By minimality the forward orbit of x is dense, so every word of X occurs in $x_{[0, \infty)}$, and $x_{[-\ell, \ell]}$ recurs at arbitrarily large positions. So there is m with $x_{[m-\ell, m+\ell]} = x_{[-\ell, \ell]}$ such that every word of length $2\ell + 1$ of X occurs in $x_{[0, m)}$. The m -periodic sequence y_ℓ that agrees with x on $[0, m)$ then agrees with x on $[-\ell, m + \ell]$. Every window of length $2\ell + 1$ of y_ℓ is a translate of one centered in $[0, m)$, so y_ℓ has the same words of length $2\ell + 1$ as X . Let N_ℓ be its least period, and on $\mathbb{F}_2^{\mathbb{Z}/N_\ell\mathbb{Z}}$ let $P\delta_t = \delta_{t+1}$. For f depending only on coordinates in $[-\ell, \ell]$ let $D_\ell(f)\delta_t = f(T^t y_\ell)\delta_t$, where $f(T^t y_\ell)$ is the value of f at any point of X that agrees with $T^t y_\ell$ on $[-\ell, \ell]$. Put $\varphi_\ell(\sum_j f_j u^j) = \sum_j D_\ell(f_j)P^j$ when every f_j is of this kind, and $\varphi_\ell = 0$ otherwise. Since $PD_\ell(f)P^{-1} = D_\ell(f \circ T^{-1})$ whenever f depends only on coordinates in $[-\ell + 1, \ell - 1]$, for fixed $r, r' \in R$ the identities $\varphi_\ell(r + r') = \varphi_\ell(r) + \varphi_\ell(r')$ and $\varphi_\ell(rr') = \varphi_\ell(r)\varphi_\ell(r')$ hold for large ℓ , and $\varphi_\ell(1) = I$. The shifts $T^t y_\ell$, $0 \leq t < N_\ell$, are distinct, so products of the $P^i D_\ell(e_a)P^{-i}$ give every diagonal matrix unit, and with P every matrix unit. Therefore $\varphi_\ell(S)$ generates $M_{N_\ell}(\mathbb{F}_2)$, and Theorem 2 applies. $\square$

Sharpness over $\mathbb{Z}$. For a homeomorphism T of a Cantor set X , the group $\mathrm{EL}_3(\mathrm{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z})$ is simple and Kazhdan if and only if (X, T) is conjugate to a minimal subshift. One direction is Theorem 1. Conversely, a Kazhdan group is finitely generated [3, Theorem 1.3.1], and the entries of finitely many generators and their inverses generate the ring, as $e_{12}(r)$ has the entry r . So the ring is generated by finitely many $\sum_j f_j u^j$, the translates of these f_j generate the clopen sets, and the itinerary map to the atoms they generate is a conjugacy onto a subshift. If a nonempty proper closed set Y is invariant, restriction maps the group onto $\mathrm{EL}_3(\mathrm{LC}(Y, \mathbb{F}_2) \rtimes_T \mathbb{Z})$ with a kernel that contains $e_{12}(e_U)$ for a nonempty clopen set $U \subseteq X \setminus Y$ but not $e_{12}(1)$, so the group is not simple.

An example. Let X be the Fibonacci subshift $X_{(3-\sqrt{5})/2}$ of Corollary 5. Its words of length 3 are 001, 010, 100 and 101. These are also the words of length 3 of the 5-periodic sequence y with $y_{[0,5)} = 01001$, and no smaller period is possible. With $\ell = 1$ and y in place of y_ℓ we get

$N = 5$ and the model of Figure 1(a): $\varphi(e_0) = I - \varphi(e_1)$ and

$$\varphi(u) = P = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & 1 \\ 1 & \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & \cdot \\ \cdot & \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & 1 & \cdot \end{pmatrix}, \quad \varphi(e_1) = D_1(e_1) = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & 1 \end{pmatrix},$$

with rows and columns indexed by $t = 0, \dots, 4$ and zeros shown as dots. The windows of y centered at 1 and at 4 are both 010, so every $D_1(f)$ has equal entries at these positions. But $P^3 D_1(e_1) P^{-3}$ is diagonal with entry y_{t-3} at t , so $D_1(e_1) P^3 D_1(e_1) P^{-3} = E_{44}$. Then $P^{t-4} E_{44} P^{4-t} = E_{tt}$ and $P^{t'-t} E_{tt} = E_{t't}$ give every matrix unit. So $\varphi(S)$ generates $M_5(\mathbb{F}_2)$, and the $e_{ij}(\varphi(s))$ with $s \in S$ generate $\text{SL}_{15}(\mathbb{F}_2)$, as in Section 1.

3. LEF GROUPS

Kionke and Schesler asked which groups embed in finitely generated simple amenable groups [22]. Corollary 3 answers the analogous question for simple Kazhdan groups that are limits of finite simple groups. Its last statement is the analogue for LEF groups of the theorem of Boone and Higman that a finitely generated group has solvable word problem if and only if it embeds in a simple subgroup of a finitely presented group [4], in Thompson's form, where the simple group is finitely generated and has solvable word problem [39].

Corollary 3. *A finitely generated group is LEF if and only if it is a subgroup of an infinite, finitely generated, simple Kazhdan group that is an expander limit of finite simple groups $\text{SL}_N(\mathbb{F}_2)$. For every finitely generated LEF group Γ there is such a group containing Γ whose word problem has the Turing degree of the word problem of Γ . So Γ has solvable word problem if and only if it is a subgroup of such a group with solvable word problem.*

Corollary 4. *Every countable LEF group is a subgroup of such a group. The members of a set of finitely generated groups embed in a single such group if and only if they are LEF and represent countably many isomorphism classes. So one such group contains every recursively presented finitely generated LEF group, for instance every finitely presented residually finite group and every G_Y with $L(Y)$ recursive (Corollary 5).*

Simple hosts with only one of property (T) and LEF were known. Without property (T), Kionke and Schesler proved that every finitely generated residually finite group embeds in a finitely generated simple LEF group [22, Theorem 1.2], so with Wilson's theorem [42] one simple LEF group contains every finitely presented residually finite group. Without LEF, every countable group embeds in a finitely generated simple group [13, 35], and even in a simple quotient of a torsion-free hyperbolic Kazhdan group [9, Theorem 1.5], so one simple Kazhdan group contains Higman's universal finitely presented group [19].

We embed a countable LEF group in the derived subgroup of a 2-generated LEF group Δ (Lemma A), and then $[\Delta, \Delta]$ in a group of Theorem 2(b) built from a lamplighter action of Δ (Lemma B). Lemma C keeps the Turing degree of the word problem in both steps.

Lemma A. *Every countable LEF group is a subgroup of $[\Delta, \Delta]$ for an infinite LEF group Δ generated by two elements.*

Proof. Let $\Gamma = \{\gamma_1, \gamma_2, \dots\}$ be a countable LEF group; replacing Γ by $\Gamma \times \mathbb{Z}$, we may assume that Γ is infinite. There are finite groups Q_k and maps $\mu_k: \Gamma \rightarrow Q_k$ that are injective on $\{1, \gamma_1, \dots, \gamma_k\}$ and preserve the products that stay in this set. On $Q_k \times \mathbb{Z}/m_k$, with $m_k = |Q_k|$, let t_k add 1 to the second coordinate, and let f_{jk} multiply the first coordinate of (x, i) from

the left by $\mu_k(\gamma_j)^i$, which is well defined because the order of $\mu_k(\gamma_j)$ divides m_k . Then $[f_{jk}, t_k]$ multiplies the first coordinate by $\mu_k(\gamma_j)$, as in the embedding of B. H. Neumann and H. Neumann [29]. On $Q_k \times \mathbb{Z}/m_k \times \mathbb{Z}/4^{k+1}$ let b_k add 1 to the last coordinate, and let a_k act on the level i , the set of points with last coordinate i , by f_{jk} if $i = 2^{2j-1}$ and by t_k if $i = 4^j$, for $j \leq k$, and trivially on the other levels. Then $b_k^{-n} a_k b_k^n$ acts on the level i as a_k acts on the level $i + n$. As distinct ordered pairs of distinct powers of 2 have distinct nonzero differences, all of absolute value below a quarter of the modulus, the two conjugates in $w_j = [b^{-2^{2j-1}} a b^{2^{2j-1}}, b^{-4^j} a b^{4^j}]$, evaluated at (a_k, b_k) , cannot both act nontrivially on a level other than 0. So $w_j(a_k, b_k)$ acts by $[f_{jk}, t_k]$ on the level 0 and trivially on the other levels, for $j \leq k$. In the algebraic ultraproduct over ω of the symmetric groups of these sets, sending γ to the class of the permutations multiplying the first coordinate on the level 0 by $\mu_k(\gamma)$ and acting trivially elsewhere is a homomorphism, injective as $\mu_k(\gamma) \neq 1$ for $\gamma \neq 1$ and large k . It sends γ_j to $w_j(a, b)$, where $a = (a_k)_k$ and $b = (b_k)_k$, so $\Gamma \leq [\Delta, \Delta]$ for the group Δ generated by a and b . Finitely many equations and inequalities that hold in the ultraproduct hold in some coordinate, so Δ is LEF, and Δ is infinite because it contains Γ . $\square$

Lemma B. *For a finitely generated group Δ , let Δ act on $\Omega = \mathbb{F}_2^\Delta$ by $(\delta x)(\eta) = x(\delta^{-1}\eta)$, let Λ be the group of maps $x \mapsto \delta x + c$ with $\delta \in \Delta$ and c finitely supported, and put $R_\Delta = \text{LC}(\Omega, \mathbb{F}_2) \rtimes \Lambda$ and $G_\Delta = \text{EL}_3(R_\Delta)$. If Δ is infinite and LEF, then G_Δ is an infinite, finitely generated, simple Kazhdan group and an expander limit of finite simple groups $\text{SL}_{3N}(\mathbb{F}_2)$, and $\xi \mapsto \text{diag}(u_\xi, 1, 1)$ embeds $[\Lambda, \Lambda]$, which contains $[\Delta, \Delta]$, in G_Δ .*

Proof. The group Λ is generated by Δ and the map z adding 1 at e , and its orbits contain the dense sets $x + \bigoplus_\Delta \mathbb{F}_2$. If $(\delta, c) \neq (e, 0)$, the fixed points of $x \mapsto \delta x + c$ satisfy $x(\eta) = x(\delta^{-1}\eta) + c(\eta)$ for all η . In every cylinder over a finite set W this fails at some point, for $\eta \in \text{supp } c$ if $\delta = e$ and for $\eta \notin W \cup \delta W \cup \text{supp } c$ if $\delta \neq e$. So the action is minimal and topologically free. The ring R_Δ is generated by 1, the units $u_\xi^{\pm 1}$ for ξ in a finite generating set of Λ , and the indicator e_U of $U = \{x : x(e) = 1\}$.

Let $\sigma_n : \Delta \rightarrow Q'_n$ be finite models of Δ that are injective on the ball of radius n and preserve the products that stay in it, with $\sigma_n(e) = e$ and Q'_n generated by the images of the generators. To $\xi : x \mapsto \delta x + c$ in Λ assign the permutation p_ξ of $Y_n = \mathbb{F}_2^{Q'_n}$ given by $y \mapsto \sigma_n(\delta)y + \sum_{\eta \in \text{supp } c} 1_{\sigma_n(\eta)}$, where $(qy)(q') = y(q^{-1}q')$, and attach to y the point $y \circ \sigma_n$ of Ω . For fixed $\xi, \xi' \in \Lambda$ and large n we have $p_\xi p_{\xi'} = p_{\xi\xi'}$, and for every finite $W \subseteq \Delta$ and large n the attached points satisfy $(p_\xi y) \circ \sigma_n = \xi(y \circ \sigma_n)$ on W for all y , since σ_n respects the finitely many products and inequalities involved. Let $D_n(f)$ be the diagonal matrix on $\mathbb{F}_2^{Y_n}$ with entry $f(y \circ \sigma_n)$ at y , let P_ξ be the permutation matrix of p_ξ , and put $\varphi_n(\sum_\xi f_\xi u_\xi) = \sum_\xi D_n(f_\xi) P_\xi$. For fixed f and ξ and large n ,

$$P_\xi D_n(f) P_\xi^{-1} = D_n(f \circ \xi^{-1}),$$

so for fixed $r, r' \in R_\Delta$ and large n , φ_n is additive and multiplicative on r, r' , and $\varphi_n(1) = I$. The conjugates of $D_n(e_U)$ by products of the P_ξ with ξ in the generating set are the coordinate functions of Y_n , and the conjugates of P_z are the translations by the 1_q , so the images of the generators generate $M_{2|Q'_n|}(\mathbb{F}_2)$. Therefore Theorem 2 applies to R_Δ .

Finally, $\xi \mapsto \text{diag}(u_\xi, 1, 1)$ is an injective homomorphism $\Lambda \rightarrow \text{GL}_3(R_\Delta)$. It maps $[\Lambda, \Lambda]$ into G_Δ , since over $\mathbb{F}_2$ every unit v satisfies $\text{diag}(v, v^{-1}, 1) = e_{12}(v) e_{21}(v^{-1}) e_{12}(v) e_{12}(1) e_{21}(1) e_{12}(1)$, and the product of these matrices for $v = v_1, v_2$ and $(v_2 v_1)^{-1}$ is $\text{diag}(v_1 v_2 v_1^{-1} v_2^{-1}, 1, 1)$. As $\Delta \leq \Lambda$, the group $[\Lambda, \Lambda]$ contains $[\Delta, \Delta]$. $\square$

Proof of Corollary 4 and of the first statement of Corollary 3. Limits of finite groups in the space of marked groups are LEF, and so are their subgroups. Conversely, by Lemmas A and B every countable LEF group is a subgroup of a group G_Δ as in the corollaries. For the second statement of Corollary 4, apply the first to the restricted direct sum of representatives of countably many isomorphism classes of finitely generated LEF groups, which is a countable LEF group, as finite subsets lie in finite products. Conversely, a countable group has countably many finitely generated subgroups. The last statement of Corollary 4 follows, as there are countably many recursive presentations. $\square$

The word problem of the host.

Lemma C. (a) *Every finitely generated LEF group Γ is a subgroup of $[\Delta, \Delta]$ for an infinite finitely generated LEF group Δ whose word problem is Turing reducible to that of Γ .* (b) *The word problem of G_Δ is Turing reducible to that of Δ .*

Proof. Let Γ be a finitely generated LEF group, generated by a finite set E . We use a half-line form of the Neumann–Neumann construction, which Darbinyan used to preserve solvable word problems [10]. On $\Gamma \times \mathbb{Z}$, whose second coordinate is the level, let β raise the level by 1, and let h_γ multiply the first coordinate from the left by γ on the levels $m \geq 0$. Then $[\beta, h_\gamma]$ multiplies the level 0 by γ^{-1} , so $\Delta = \langle \beta, h_\gamma : \gamma \in E \rangle$ is infinite and $\Gamma \leq [\Delta, \Delta]$. Let $\lambda \geq 1$. An element of Δ of length at most λ is $f\beta^\ell$ with $|\ell| \leq \lambda$, where f multiplies the level m from the left by an element $f(m)$ of length at most λ , with $f(m) = 1$ for $m < -\lambda$ and $f(m) = f(\lambda)$ for $m > \lambda$. Let ψ map the ball of radius 2λ of Γ injectively into a finite group Q , preserving the products that stay in it, and let θ be the isomorphism from $\{v \in Q^{[-3\lambda, 3\lambda]} : v_{3\lambda-1} = v_{3\lambda}\}$ onto $\{v \in Q^{[-3\lambda, 3\lambda]} : v_{-3\lambda} = 1\}$ given by $\theta(v)_{-3\lambda} = 1$ and $\theta(v)_m = v_{m-1}$ for $m > -3\lambda$. The bijection κ of $Q^{[-3\lambda, 3\lambda]}$ given by $\kappa(x)_{-3\lambda} = x_{3\lambda-1}^{-1}x_{3\lambda}$ and $\kappa(x)_m = x_{m-1}$ for $m > -3\lambda$ satisfies $\kappa(vx) = \theta(v)\kappa(x)$ for v in the domain of θ , as $v_{3\lambda-1} = v_{3\lambda}$ cancels in the coordinate -3λ . Let each $v \in Q^{[-3\lambda, 3\lambda]}$ act on $Q^{[-3\lambda, 3\lambda]} \times \mathbb{Z}/5\lambda$ by left multiplication on the first coordinate, and put $\zeta(x, i) = (\kappa(x), i + 1)$, so that $\zeta v \zeta^{-1} = \theta(v)$ on the domain of θ . As f is trivial below $-\lambda$ and constant from λ on, $\zeta^\ell(\psi(f(m)))_m \zeta^{-\ell} = (\psi(f(m - \ell)))_m$ for $|\ell| \leq \lambda$. So $f\beta^\ell \mapsto (\psi(f(m)))_{|m| \leq 3\lambda} \zeta^\ell$ preserves the products that stay in the ball of radius λ , and it is injective there, as $5\lambda > 2\lambda$ and f is determined by its values for $|m| \leq 3\lambda$. Therefore Δ is LEF.

From a word in the $\beta^{\pm 1}$ and $h_\gamma^{\pm 1}$ one computes the exponent sum ℓ of β , the levels where f changes, namely the exponent sums of β before the letters $h_\gamma^{\pm 1}$, and the values of f there as words in E . It is trivial if and only if $\ell = 0$ and these values are trivial in Γ , so the word problem of Δ is Turing reducible to that of Γ . This proves (a).

For (b), an element $x \mapsto \delta x + c$ of Λ is given by a word for δ and words for the points of $\text{supp } c$, so an oracle for the word problem of Δ decides equality in Λ and computes the values of c . A word in the generators of G_Δ multiplies out to a matrix with entries $\sum_\xi p_\xi u_\xi$, where each p_ξ is a polynomial over $\mathbb{F}_2$ in finitely many coordinates $x(\delta)$, as $u_\xi e_U u_\xi^{-1}$ is the indicator of $\{x : x(\delta) = 1 + c(\delta)\}$ for $\xi : x \mapsto \delta x + c$. After collecting equal ξ and equal coordinates, the word is trivial if and only if every coefficient of its difference from I_3 vanishes at every assignment of its finitely many variables, as Ω is the full shift. So the word problem of G_Δ is Turing reducible to that of Δ . $\square$

Proof of the second statement of Corollary 3. Let Γ be a finitely generated LEF group. By Lemmas C and B, Γ is a subgroup of a group G_Δ as in Corollary 3 whose word problem is Turing reducible to that of Γ . Conversely, a word in the generators of Γ becomes a word in the generators of G_Δ by substitution, and it is trivial in Γ if and only if its image is trivial in G_Δ , so the word problem of Γ reduces to that of G_Δ . The last statement follows, as finitely

generated subgroups of groups with solvable word problem have solvable word problem and LEF passes to subgroups. $\square$

4. WORD PROBLEMS

Corollary 5. *The word problem of G_X has the Turing degree of the language $L(X)$ of X , the set of words occurring in points of X , so G_X has solvable word problem if and only if $L(X)$ is recursive, as for the Fibonacci subshift $X_{(3-\sqrt{5})/2}$ below. Every Turing degree is the degree of the word problem of some G_X , and there is a family of continuum many groups G_X , none of which embeds in another.*

Proof. Multiplying out a word in the generators over $\text{LC}(A^{\mathbb{Z}}, \mathbb{F}_2) \rtimes \mathbb{Z}$, which maps onto R , with the rule $uf = (f \circ T^{-1})u$ gives a matrix with entries $\sum_j f_j u^j$, where each f_j is given by a table on the words of some length. The word is trivial in G_X if and only if the tables of its difference from I_3 vanish on $L(X)$, so $L(X)$ computes the word problem. Conversely, by (1) a word for $e_{12}(\prod_{t < m} u^{-t} e_{v_t} u^t)$ is computable from $v = v_0 \cdots v_{m-1}$, and it is trivial if and only if $v \notin L(X)$.

For irrational $\alpha \in (0, 1)$ let X_α be the infinite minimal Sturmian subshift, the closure of the codings $c(\theta)$, $\theta \in [0, 1)$, with $c(\theta)_t = \lfloor \theta + (t+1)\alpha \rfloor - \lfloor \theta + t\alpha \rfloor$, which is 1 exactly for θ in the arc $[-(t+1)\alpha, -t\alpha)$ modulo 1 [28, 18], [24, Chapter 2]. Passing to the closure adds no words, so the words of length m of X_α are the constant values of $c(\theta)_{[0, m]}$ on the arcs between the distinct points $-j\alpha \bmod 1$, $0 \leq j \leq m$, and α computes $L(X_\alpha)$. Such a word has $\lfloor \theta + m\alpha \rfloor$ ones, within 1 of $m\alpha$, so $L(X_\alpha)$ computes α . Every $D \subseteq \mathbb{N}$ has the degree of $[0; 1 + \chi_D(0), 1 + \chi_D(1), \dots]$, since each digit is decided by one strict comparison with a rational computed from earlier digits. If a finitely generated group H embeds in a finitely generated group G , a word for H becomes a word for G by substituting words in the generators of G for the generators of H , so the word problem of H reduces to that of G . By a construction of Sacks, the Turing degrees contain an antichain of size continuum [30, Chapter V], and the groups G_{X_α} with α of these degrees are as claimed. $\square$

For derived topological full groups, Grigorchuk and Medynets proved that the word problem is solvable if and only if $L(X)$ is recursive [16, Theorem 1.1(3)].

5. THE FINITELY PRESENTED CASE

No group in Theorem 2(b) is finitely presented: a finitely presented LEF group is residually finite [37, 40], and an infinite simple group is not. Infinite finitely presented simple Kazhdan groups exist [6], and whether one of them is sofic, or at least hyperlinear, remains open. A sofic example would not be LEF, and it would answer Open problem 6.1 of Alekseev and Thom [1], which asks for finitely presented sofic groups with property (T) that are not residually finite.

ORIGIN AND AUTHORSHIP

Claude (Anthropic) found the construction and original proofs under the author's direction on September 12, 2026, and wrote the manuscript. Proof records are in the [project repository](#).

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